PES University — Department of Computer Science & Engineering

Lab 3: Software Requirements Specification (SRS) & WBS

Problem Statement #55: Book Club Reading Challenge & Discussion Portal | **Domain:** Media, Events & Community
Student: Sushruth S | **SRN:** PES1UG24CS486 | **Section:** 4-H

1. IEEE 830 Software Requirements Specification (SRS) Overview

1.1 Product Purpose & Scope: The Book Club Reading Challenge & Discussion Portal is a collaborative platform providing gamified reading tracking, chapter discussion threads with interactive spoiler protection, role-based moderation for Discussion Leads, and democratic monthly book selection polls.
1.2 User Classes & Roles: *Club Member* (tracks pages, reads/replies, votes once/cycle), *Discussion Lead* (creates threads, pins prompts, deletes inappropriate posts), and *Backend System* (session authentication, Redis caching, atomic ballot writes).
1.3 Key Functional Capabilities:
• **FR-001 (Spoiler Masking):** Hides [spoiler] comments behind CSS blur filter; sanitizes search previews.
• **FR-002 (Reading Goals):** Validates integer page logs ($0 < ext{pages} \le ext{remaining}$); updates completion % and streaks.
• **FR-003 (Moderation & RBAC):** Discussion Leads moderate content; regular members restricted (403 Forbidden).
• **FR-004 (Monthly Poll):** 1 vote per member; atomic transaction prevents duplicate submissions (409 Conflict).
• **FR-005 (Badge Engine):** Awards milestone achievement badges asynchronously upon reaching goal thresholds.

2. Non-Functional Quality Requirements

• **NFR-001 (Performance):** Sub-200ms latency for 95% of poll voting requests under 1,000 concurrent users via Redis caching.
• **NFR-002 (Security & Availability):** 99.9% uptime (≤ 43.8 mins downtime/month); AES-256 encryption at rest; TLS 1.3 in transit.
• **NFR-003 (Usability):** Responsive viewport support (375px to 1920px); WCAG 2.1 Level AA accessibility compliant.
• **NFR-004 (Data Integrity):** PostgreSQL composite unique key on (user_id, poll_id) eliminates duplicate ballots.

3. Work Breakdown Structure (WBS) & Sprint Allocation

The engineering lifecycle is decomposed into 6 structured phases. Core functional modules are estimated using the Fibonacci scale totaling **29 Story Points** (matching the Jira Scrum backlog).

WP ID	Work Package / Deliverable	Phase	Pts	Predecessors	Status
WP-1.1	Stakeholder & Domain Modeling	Phase 1: Inception	2	None	**Done**
WP-1.2	FR & NFR Specification Table	Phase 1: Inception	3	WP-1.1	**Done**
WP-1.3	Requirements Traceability Matrix (RTM)	Phase 1: Inception	3	WP-1.2	**Done**
WP-1.4	UML Use Case Diagram & Flow Spec	Phase 1: Inception	5	WP-1.2	**Done**
WP-2.1	Jira Kanban & Scrum Workspace Setup	Phase 2: Agile Jira	3	WP-1.3	**Done**
WP-2.2	Product Backlog (29 Story Points)	Phase 2: Agile Jira	3	WP-2.1	**Done**
WP-2.3	Sprint 1 Simulation & Burndown Chart	Phase 2: Agile Jira	5	WP-2.2	**Done**
WP-2.4	Defect Logging & Reports (BTBSC-1..4)	Phase 2: Agile Jira	3	WP-2.3	**Done**
WP-3.1	4-Tier Layered Architecture Design	Phase 3: Architecture	5	WP-1.4	**Done**
WP-4.1	Copilot Prompting & Engine Implementation	Phase 4: Engineering	5	WP-3.1	**Done**
WP-4.2	Jira Bug Remediation (BTBSC-1 to 4)	Phase 4: Engineering	5	WP-4.1	**Done**
WP-5.1	Software Testing Practice (4 Test Cases)	Phase 5: Testing	3	WP-4.2	**Done**
WP-5.2	Vibe Coding Bug Fix & Retest Suite	Phase 5: Testing	3	WP-5.1	**Done**
WP-6.1	Master README & Multi-Folder Release	Phase 6: Release	3	WP-5.2	**Done**